

Visual Predictive Checks for Real-World Data using Propensity-Score Matching

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Background

- We reported recently that visual predictive checks (VPC) cannot be used reliably when real world data (RWD) or data from adaptive trials are analyzed. [1]
- Prediction-correction (pcVPC) may help in some cases (e.g. changes only in dose amount), however, it still provides an incorrect diagnosis for most other cases.
- Here we present a potential solution to this problem, in the form of **propensity-score-matched VPCs** (pmVPC).
- The pmVPC requires a modification in the simulation step of the VPC: instead of randomly assigning drawn parameter sets to individual dataset designs from observed subjects, they are now matched based on the individual MAP estimates from the observed data (EBEs).

pmVPC simulation algorithm

pmVPC simulations are implemented as follows:
1. Get MAP estimates (EBE's) from fitted model (for all n individuals).

Then, for each of m simulation datasets in the VPC (e.g. $m=500$):
2. Draw n sets of individual model parameters from Ω .
3. Propensity-match each set of model parameters to an observed subjects EBE's (without replacement).
4. Simulate n subjects using the matched subject's design.

Results

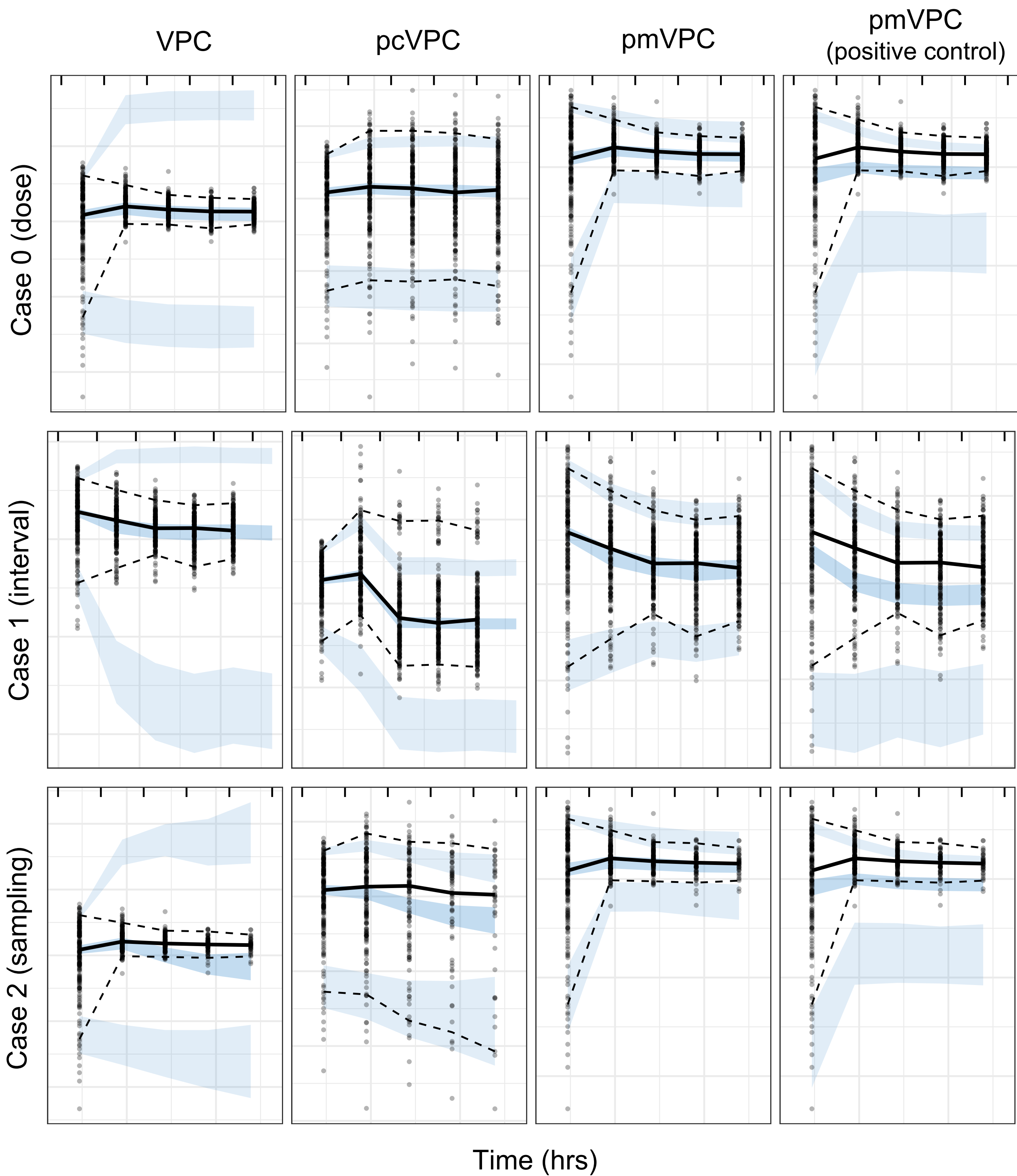


Figure 1. VPC, pcVPC, pmVPC, and positive control pmVPC for scenarios of dose-, interval- or sampling adaptation (case 3 left out, similar results). As reported in [2], the pcVPC can correct cases of dose adaptation. However, pcVPC cannot be used when interval- or sampling adaptations are in the data. The pmVPC did not show misspecification when not present, and in contrast to the pcVPC also presents data on the original scale. The pmVPC positive controls were able to detect misspecification in both scenarios (only scenario 1 shown).

Objectives

To investigate the ability of pmVPCs to overcome the limitations of VPCs and pcVPCs for RWD and adaptive trials.

Methods

- Four cases of VPCs for RWD were investigated based on simulated data from a known true model (1 cmt IV, linear elim):
 - **Case 0:** *dose* adaptation (as in [2]).
 - **Case 1:** *interval* adaptation.
 - **Case 2/3:** *sampling*-schedule adaptation.
- Positive control scenarios used misspecified models (**scenario 1:** population CL +20%, **scenario 2:** linear clearance but true model Michaelis-Menten elimination).
- Propensity-score matching was performed using the MatchIt package in R [3], based on individual estimates for CL and V, and VPCs were made using the vpc package [4].

Intuition

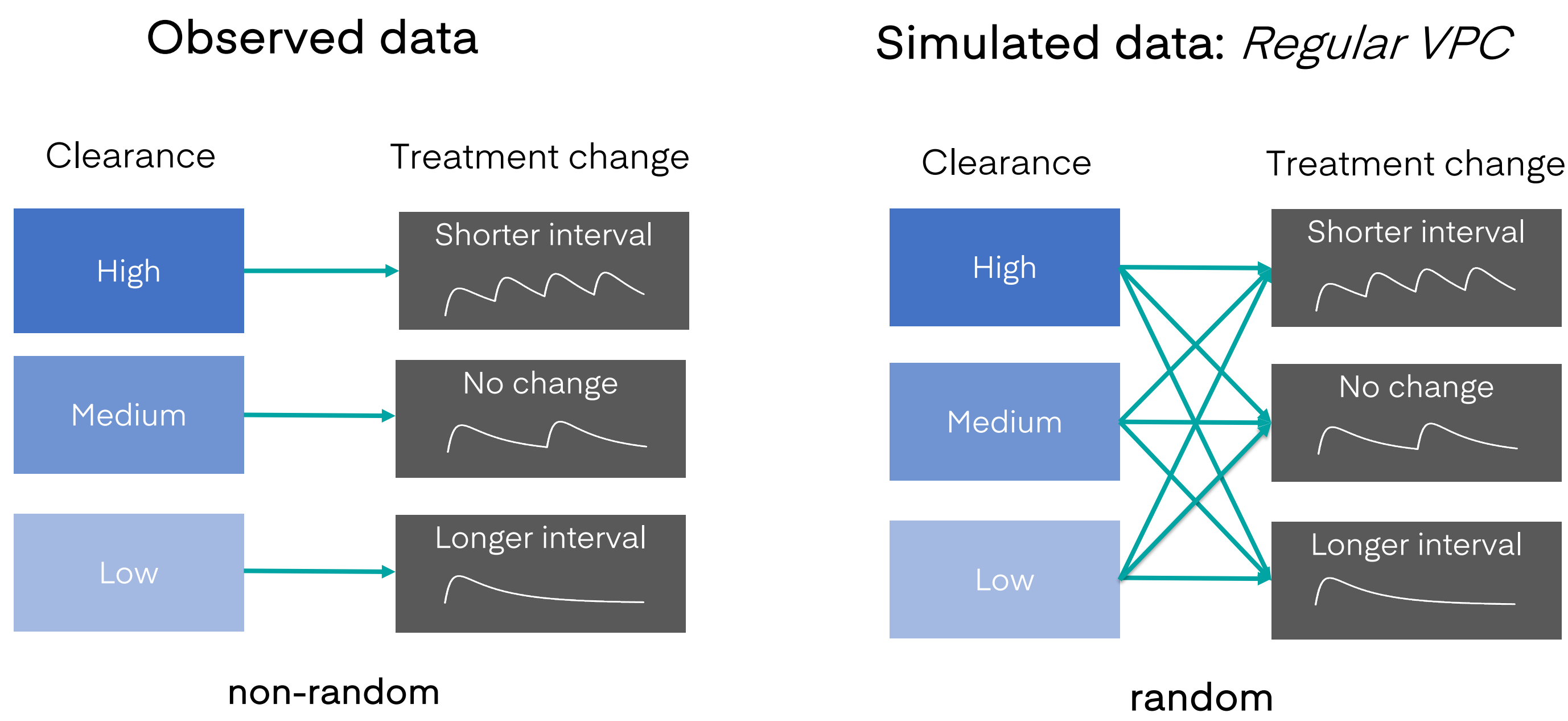
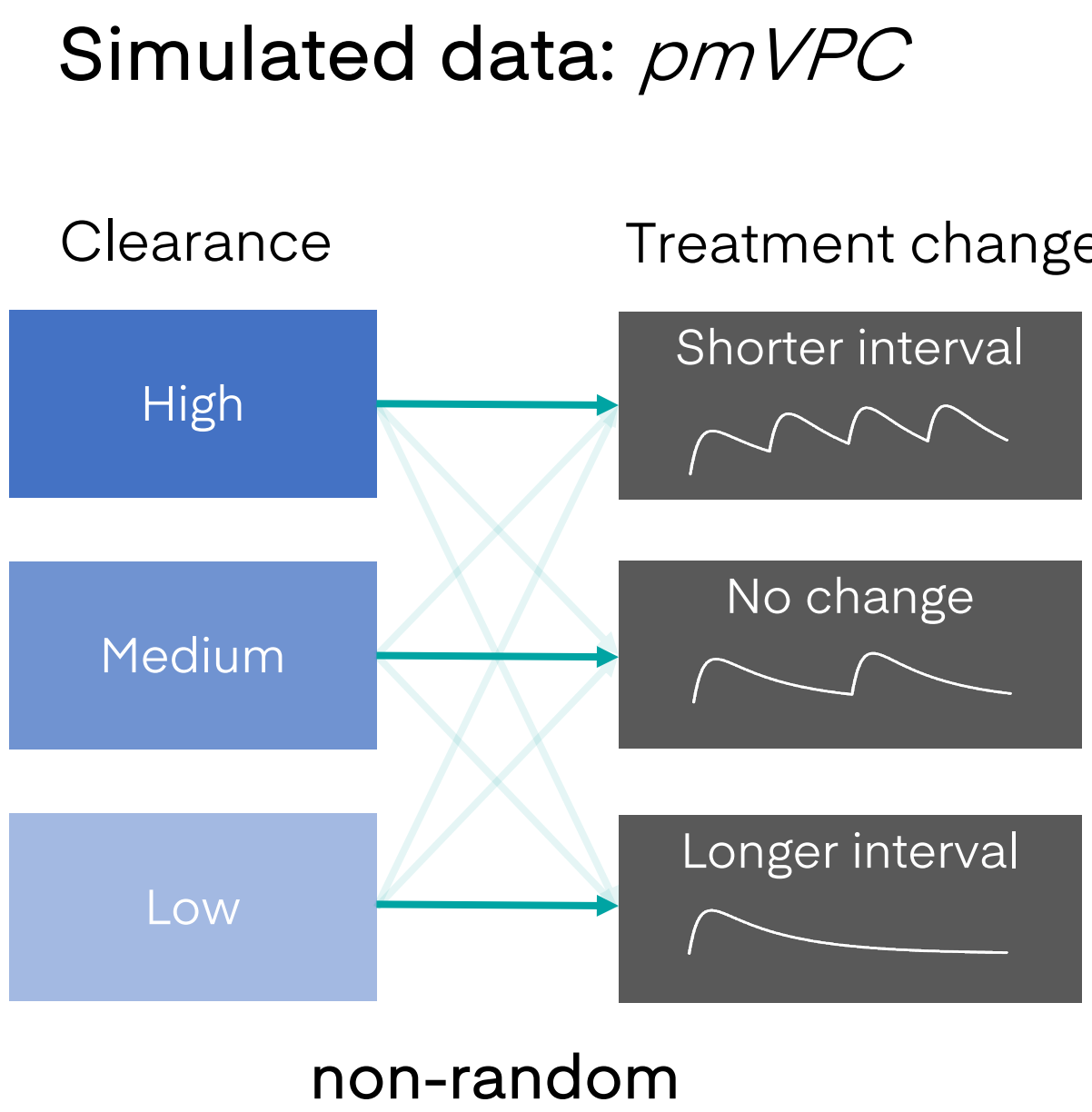


Figure 2. Visualization of mechanism by which the pmVPC enables correction of simulated data when there are biasing features in the observed data. This example shows patients with a higher CL switching to shorter intervals and patients with lower CL to a shorter interval (case 1).

- In a regular VPC, draws are **fully random**, resulting in disconnect between the subject's PK parameter and the data design.
- In the pmVPC simulations, the **correlation** between EBE and a subject's data design is restored by matching the drawn parameter sets to the observed subject.



Conclusions

- In the presence of treatment adaptation as seen in RWD, the pmVPC is able to correctly diagnose true and misspecified models, in contrast to regular and pcVPCs.
- In combination with other novel VPC methodology [5,6] the pmVPC offers an improved toolbox for model evaluation for complex datasets.

References

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2. Bergstrand M and Karlsson MO. AAPS J. 2011
3. <https://github.com/kosukeimai/MatchIt>
4. <https://github.com/ronkeizer/vpc>
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